P04

Self-Organizing Traffic Flow in Distributed Topologies

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Domain DOM_NET

Abstract

Decentralized control in physical and digital traffic networks can prevent phantom jams and buffer overflows. Local queue-length feedback coupled with stochastic boundary conditions leads to stable macro-scale flows without global schedule arbitration.

Abstract quoted from D. Helbing et al., Nature (1998).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?